WEEK 14 • IS321 ENTERPRISE ARCHITECTURE

AI in Modern Enterprise Architecture

How AI Is Reshaping Modeling, Governance, and the EA Repository

TODAY'S ROADMAP

What We'll Cover

01	**Why AI in EA, Why Now**	What changed, and why architects are paying attention
02	**AI-Assisted Architecture Modeling**	Drafting Current Architecture faster, with the architect still verifying
03	**AI-Supported Governance & Smart Repositories**	Continuous compliance scanning, plus semantic search across the EA Repository
04	**Risks, Limitations & Guardrails**	What AI gets wrong, and how PSU policy keeps a human in the loop
05	**Worked Example & Course Recap**	Amanah Bank end to end, and how AI threads through everything covered this semester

FOUNDATIONS

Why AI in EA, Why Now

Bernard's EA³ Cube and TOGAF's ADM both predate today's AI tooling. Neither framework changes — but three shifts explain why AI is being layered onto EA practice right now, across both frameworks, not replacing either.



Documentation Volume

Enterprises now run thousands of applications across dozens of Segments — far more than any team can document and keep current by hand.



Mature AI Tooling

Large language models and pattern-matching tools are now reliable enough to draft first-pass documentation from existing sources, not just summarize it.



Faster Change Cycles

Systems change faster than manual architecture-review cycles can track — AI closes the gap between a change happening and the repository reflecting it.

AI-Assisted Architecture Modeling

Building out the Cube's cells by hand — documenting Components, Current Architecture, and Future Architecture at every Level × Segment intersection — is exactly the kind of labor-intensive documentation AI modeling tools can accelerate. AI can draft a first-pass Current Architecture model directly from existing documentation, then flag likely gaps for the architect to verify.

EXAMPLE

An architect uploads Amanah Bank's existing system documentation; an AI modeling tool drafts a first-pass Current Architecture entry for the Retail Banking segment (per the syllabus's approved ArchiMate-compatible tooling), which the architect then reviews rather than building from scratch.

AI-Supported Architecture Governance

The Security thread (from last week's EA³ Cube session) is a governance concern that runs through every level. AI-supported governance tools can continuously scan project documentation and code repositories for drift from approved standards — catching non-conformance early instead of waiting for a scheduled compliance review.

EXAMPLE

An AI governance tool flags that a newly deployed service in the Retail Banking segment doesn't match the Standards thread's approved data-storage requirement — surfacing the issue during development rather than at the next scheduled architecture review.

Smart EA Repositories



Week 5's EA Repository (and TOGAF's Architecture Repository, Week 12) is traditionally browsed by structure or searched by keyword. An AI-enabled “smart” repository adds semantic search and automatic cross-referencing across Segments — finding related artifacts a keyword search across thousands of cube cells would miss.

EXAMPLE

An architect asks a smart repository, “Which other Segments at Amanah Bank have already built an API gateway?” and receives two matches from Commercial Banking and Wealth Management — connections a keyword search across separate segment repositories would likely miss.

A NECESSARY CAUTION

Risks and Limitations of AI in EA

AI accelerates EA work — it does not replace architectural judgment. Three limitations worth remembering:



Hallucinated Documentation

An AI model can draft a plausible-looking Current Architecture entry that does not match reality — always verify against source systems before publishing.



Governance Blind Spots

AI governance tools flag drift from documented standards — but cannot catch a standard that was never documented in the first place.



Training Data Bias

A smart repository's 'related artifacts' suggestions reflect patterns in past documentation — gaps or biases in that history carry forward.

KEEPING A HUMAN IN THE LOOP

Guardrails for Using AI in EA Work

The risks on the previous slide are exactly why PSU and this course build in explicit guardrails. Three that apply directly to your project work:

Acknowledge AI Use

Per the syllabus's AI and EA Tools policy, any AI-assisted modeling, analysis, or drafting must be disclosed — the same expectation a working architect's employer would have.

Architect Reviews, AI Drafts

Every example this week followed one pattern: AI produces a first pass; a person verifies it against source systems before it becomes official documentation.

Academic Integrity Tools Still Apply

AI-assisted submissions remain subject to PSU's academic-integrity review — using an AI tool responsibly is not the same as it doing the architectural thinking for you.

WORKED EXAMPLE

AI End-to-End: Amanah Bank's API Gateway



Trace one change through all three AI capabilities covered this week: an architect adds a new API gateway to Retail Banking. AI drafts the first-pass Current Architecture entry (modeling), a governance tool checks it against approved data-storage standards before deployment (governance), and once published, a smart repository makes it discoverable to every other Segment planning something similar (repository).

EXAMPLE

Six months later, Wealth Management's architect asks the smart repository whether anyone has built an API gateway before — and finds Retail Banking's, complete with the governance sign-off record, instead of starting from scratch.

BRIDGE ACROSS THE COURSE

How AI Threads Through the Whole Course



This isn't the first time AI has come up this semester — it's the last stop on a thread that's run since Week 3.

CONNECTS TO

- Week 3's AI-Driven Enterprise Decision Support — predictive analytics for enterprise planning.
- Week 6's AI-Based Risk Analysis — predictive risk modeling, now paired with this week's AI governance scanning.
- Week 8's AI for Business Continuity Planning and Week 10's AI-powered ERP — the same 'AI drafts, human verifies' pattern seen in every case.

CHAPTER SUMMARY

Key Takeaways

- AI-assisted modeling drafts first-pass Current Architecture documentation from existing sources — the architect verifies rather than builds from scratch.
- AI-supported governance continuously scans for drift from approved standards, catching non-conformance before the next scheduled review.
- Smart EA repositories add semantic search and cross-Segment discovery — surfacing related work a keyword search across thousands of cells would miss.
- None of this replaces architectural judgment or the syllabus's disclosure requirement: hallucinated documentation, governance blind spots, and training-data bias all require a human in the loop.
- This is the same drafts-then-verifies pattern seen everywhere AI touches this course — decision support (Week 3), risk analysis (Week 6), continuity planning (Week 8), and ERP (Week 10).
- Together, EA^3 Cube (Week 13), TOGAF (Week 12), and this week close the loop: frameworks give AI something structured to work within, not a blank page.

COMING UP NEXT

Further Reading & Next Session

Reading Material

- Bernard, S. (2012). An Introduction to Enterprise Architecture (3rd ed.) — Chapters 8–9, EA^3 Cube and TOGAF context for this week's AI content
- PSU IS321 Syllabus — AI and EA Tools policy
- Weeks 3, 6, 8, 10, 12, and 13 — every week this course touched AI or a framework AI is now layered onto

Next: Week 15

Course integration, project review, and final-exam preparation — no new content, just bringing Weeks 1–14 together. Mapped to CLO1–CLO6.

Questions?